

Beyond Homogeneous Adaptation: Decoupled Control for Sample- and Layer-Aware Test-Time Robustness in Temporally Correlated Streams

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Abstract—Test-Time Adaptation (TTA) enables pre-trained models to adapt online to distribution shifts during inference, but its self-training dynamics are fragile under temporally correlated streams. Once biased target evidence is used for adaptation, the resulting errors can accumulate over time, distort target representations, and persist unevenly across network layers. Existing methods typically mitigate this problem with confidence filtering or global parameter regularization, but these homogeneous controls overlook two forms of heterogeneity: target samples differ in whether they provide trustworthy adaptation evidence, and layers differ in whether they can safely absorb the induced updates. We propose DCF, a Decoupled Control Framework that stabilizes TTA by jointly controlling *which samples drive adaptation* and *which layers retain the resulting update*. At the sample level, DCF uses Probe-supported Sample Routing (PSR) to identify probe-supported trusted evidence via entropy and a Fourier stress response, while Routed-away Geometry Repair (RGR) reuses routed-away samples for optimal-transport-based geometry repair instead of discarding them. At the parameter level, the sample-side update is treated as only a candidate update: Curvature-aware Layer Retention (CLR) then performs layer retention, preserving plasticity in layers while suppressing updates in layers that exhibit high source mismatch. This route-adapt-retain design prevents unreliable evidence from dominating adaptation while avoiding indiscriminate rollback of useful plasticity. Extensive experiments on temporally correlated corruptions, natural shifts, and long-horizon streams demonstrate that DCF consistently improves test-time robustness and avoids the late-stage collapse of existing TTA methods. Code is available at <https://github.com/ioslide/DCF>.

Index Terms—Deep learning, test-time robustness, test-time adaptation, domain shift.

I. INTRODUCTION

Deep neural networks [1, 2] excel when training and test data are independently and identically distributed (i.i.d.), yet

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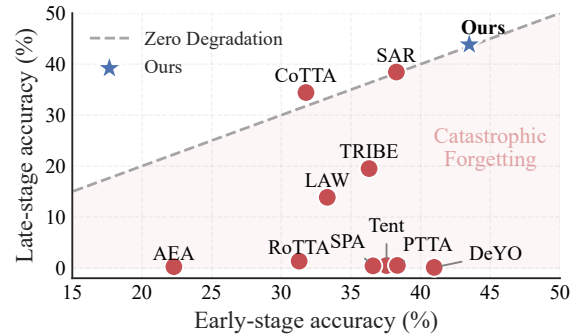


Fig. 1: **Long-horizon instability of existing TTA methods.** Each point represents a method’s average accuracy at the early stage (x-axis, Domain index 15) and late stage (y-axis, Domain index 225) of adaptation on a temporally correlated stream. Many methods that achieve strong early performance later fall below the diagonal, indicating unstable long-horizon adaptation.

their reliability degrades sharply once deployment environments deviate from the training distribution [3–5]. In practical visual recognition systems, test samples rarely arrive as randomly shuffled instances from a fixed target domain. Instead, they typically form temporally correlated streams in which sensor noise, illumination, weather, corruption type, and rendering style evolve gradually over time. Test-Time adaptation (TTA) [6, 7] addresses this setting by updating a pre-trained source model online using only unlabeled test samples, making it an appealing paradigm for source-free robustness under distribution shift.

Despite its promise, self-training dynamics in online TTA are inherently fragile under temporal correlation. Since updates rely on the model’s own predictions, local classification errors are easily reinforced. Over extended horizons, biased target evidence distorts internal representations and alters the model’s sensitivity to future samples, potentially causing catastrophic late-stage collapse. As shown in Figure 1, several TTA methods achieve competitive early-stage accuracy yet deteriorate substantially in the late stage, raising a central question:

Why does test-time adaptation under temporally correlated shifts become unstable over time?

Existing methods [6, 8–16] typically mitigate this instability through two forms of homogeneous control: 1) filtering samples by prediction confidence, which implicitly assumes that confi-

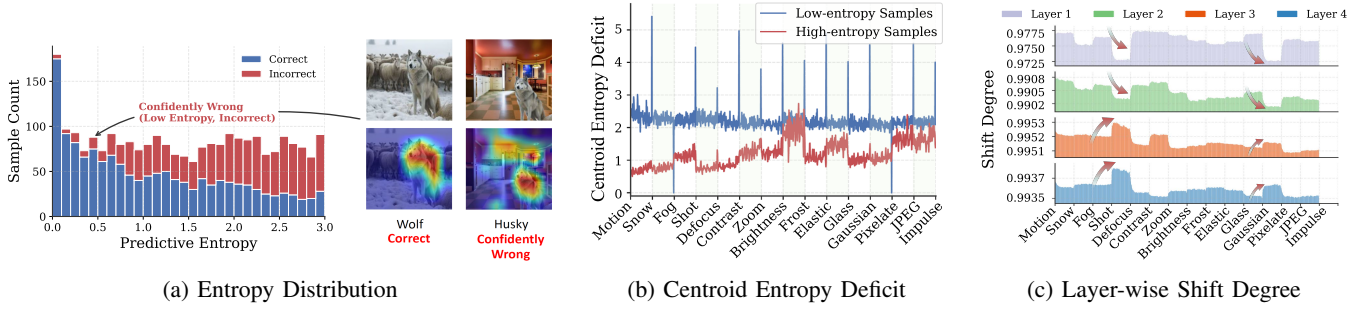


Fig. 2: **Why homogeneous TTA fails.** (a) Low entropy does not guarantee reliability: even highly confident predictions can be wrong and may be driven by spurious cues. (b) High-entropy samples are weak pseudo-label candidates, but their centroid entropy deficit indicates that they still capture shifting target geometry. Discarding them accelerates representation bias and removes useful structural anchors. (c) Layer drift, measured by KL divergence from the source model, is heterogeneous across depth, with different magnitudes and even opposite trends.

dent predictions provide reliable pseudo-label supervision, and 2) regularizing parameter movement, which implicitly assumes that adaptation-induced drift can be suppressed by a global or nearly uniform stabilization rule. Although effective in certain settings, these strategies neither fully explain nor resolve the failure pattern observed on long temporally correlated streams.

We argue that the instability of temporally correlated TTA is not merely caused by pseudo-label noise or parameter drift in isolation. Instead, it is better understood as a coupled sample-layer feedback process. At the sample level, unreliable target evidence induces biased self-training updates: a low-entropy prediction may not be reliable and may be driven by shortcut-sensitive cues, whereas high-entropy samples, although unsafe for direct pseudo-label supervision, may still encode useful structural information about the evolving target distribution. At the parameter level, such biased updates are not absorbed uniformly across network depth: some layers require plasticity to absorb benign appearance changes, whereas others suffer semantic forgetting once pushed away from the source model. Once retained in source-sensitive layers, the drift alters future predictions and biases subsequent sample selection, gradually amplifying local adaptation errors into long-horizon collapse.

This analysis reveals two overlooked forms of heterogeneity—*samples differ in whether they provide trustworthy adaptation evidence, and layers differ in whether they can safely retain the induced update*—and implies that stable TTA requires joint control over *which samples drive adaptation, how non-trusted samples are reused, and which layers retain the resulting update*.

We summarize these findings into three observations:

- **O1: Confidence is not reliability.** Low-entropy predictions can still be incorrect when they are supported by spurious or shortcut-sensitive cues. Thus, entropy alone is an insufficient criterion for selecting safe adaptation evidence (Figure 2a).
- **O2: Uncertainty is not useless.** Although high-entropy samples are unreliable for hard pseudo-label supervision, they may still contain target-domain structural information. Tracking their centroid entropy deficit shows that routed-away samples can reflect the evolving target geometry. Discarding them may accelerate representation bias and remove useful structural anchors (Figure 2b).
- **O3: Drift is depth-heterogeneous.** Adaptation-induced

drift is strongly layer-dependent, with different magnitudes and even opposite trends across layers. Therefore, a single global stabilization rule is too coarse to prevent long-horizon degradation (Figure 2c).

Motivated by these insights, we propose DCF, a Decoupled Control Framework for stable TTA in temporally correlated streams that decouples online TTA into three coordinated stages following a **Route-Adapt-Retain** principle: 1) **Route:** Probe-supported Sample Routing (PSR) partitions instances into trusted and routed-away sets by joint evaluation of predictive entropy and a Fourier-basis stress response, screening for confident, stress-responsive predictions; 2) **Adapt:** Decoupled sample-side adaptation is then conducted: trusted instances supply conservative consistency signals, while routed-away instances are repurposed by Routed-away Geometry Repair (RGR) for soft optimal-transport geometry repair rather than discarded; 3) **Retain:** The sample-side update is then passed to Curvature-aware Layer Retention (CLR), which performs closed-form, curvature-aware layer retention by balancing the candidate parameters and the source prior across layers. This decoupled control mitigates the error-amplifying sample-layer feedback loop, supporting sustained stability without sacrificing adaptive capacity.

Our main contributions are summarized as follows:

- We provide a systematic analysis of long-horizon failure in temporally correlated TTA, attributing instability to a coupled sample-layer feedback process rather than pseudo-label noise or parameter drift alone, and distilling three observations (**O1–O3**) that expose the heterogeneity of both adaptation evidence and layer-wise drift.
- We introduce DCF, a Decoupled Control Framework that implements a route-adapt-retain design by decoupling trusted-evidence routing, routed-away geometry repair, and curvature-aware layer retention.
- We conduct extensive experiments on synthetic corruptions, natural distribution shifts, label-shifted streams, mixed-severity streams, and long-horizon temporally correlated streams, demonstrating that DCF consistently improves robustness and avoids the late-stage collapse observed in strong TTA baselines.

II. RELATED WORK

We relate DCF to existing adaptation methods without and with target data, and further discuss sample reliability, target-geometry preservation, and parameter stabilization for TTA.

Adaptation without Target Data. Mitigating distribution shifts has been widely studied at training time, including domain generalization (DG) [17] and robust representation learning [18]. Beyond visual tasks, uncertainty-aware modeling has also been explored to handle evolving environmental variations in non-stationary sensor streams [19–21]. Nonetheless, such source-side designs cannot exhaustively anticipate unpredictable test shifts. In contrast, TTA directly exploits unlabeled samples observed during deployment, enabling online adaptation to evolving distributions.

Adaptation with Target Data. Target-data methods span offline unsupervised domain adaptation and online TTA.

- **Unsupervised domain adaptation (UDA).** Conventional UDA accesses labeled source and unlabeled target data to reduce cross-domain discrepancies via moment alignment [22], adversarial learning [23], pseudo-label refinement [24], or optimal transport [3]. Source-free variants recover source structure or progressively refine target representations [25]. Recent studies such as VDM-DA [4], SFOCDA [5], and TAMAN [26] further explore virtual source construction, open compound adaptation, and video-oriented alignment. However, these methods assume offline or repeatedly accessible target data, whereas TTA must adapt and predict online as test samples arrive.

- **Test-Time adaptation (TTA).** TTA adapts a pre-trained model during inference using only unlabeled test samples. Representative approaches update normalization statistics [27], minimize prediction entropy [6, 8], or exploit teacher models, memory mechanisms, and reset strategies for continual adaptation [12, 13, 28]. Test-time training [7] introduces auxiliary source-stage objectives. Recent studies extend TTA to diverse visual settings: MetaBN [29] leverages meta-trained normalization for adverse-weather video restoration, whereas CMDA [30] pairs model adaptation with diffusion-based data adaptation. In contrast, DCF requires neither meta-training nor generative translation, focusing on controlling self-training error accumulation in correlated streams. Other approaches explore sample bias [31], normalization or prior correction, and continual dynamics [9–11, 15, 32–34]. Despite these advances, self-training remains vulnerable to temporally correlated and label-imbalanced streams, where local adaptation errors can accumulate across successive updates.

Reliability- and Stability-aware TTA. Online TTA must govern how samples drive adaptation and how updates persist. EATA [8] and SAR [9] use entropy filtering, while DeYO [35] and ETAGE [16] incorporate shape- or gradient-sensitive cues. The four closely related TCSVST studies use distinct adaptation mechanisms: QED [31] performs question-conditioned VQA debiasing; MetaBN [29] meta-trains normalization for adverse-weather video restoration; CMDA [30] couples model updates with diffusion-based data adaptation; and MS-TTA [36] refines CLIP embeddings through training-free mean shift and a feature cache. DCF instead controls the coupled evolution of target evidence and model parameters in temporally correlated

recognition streams. PSR routes evidence by entropy and Fourier stress responsiveness; RGR converts routed-away samples into soft OT geometry constraints; CLR determines where the resulting candidate update persists. CoTTA [12], RoTTA [13], LAW [37], and GOLD [38] stabilize updates through weight restoration, teacher–student memory, layer-wise learning-rate modulation, or classifier-aligned subspaces. DCF evaluates candidate-update persistence separately for each layer group using current source–candidate curvature mismatch.

DCF formulates online adaptation as a coupled sample–layer feedback process, jointly controlling *which evidence drives direct supervision, how routed-away samples constrain target geometry, and where candidate updates persist*. This route–adapt–retain design interrupts the error-amplifying feedback that causes long-horizon collapse.

III. METHODOLOGY

A. Problem Setup

Test-Time Adaptation (TTA) starts from a K -class source model $f_{\theta_0} : \mathcal{X} \rightarrow \Delta^{K-1}$ trained on $\mathcal{P}_0(X, Y)$, where $f_{\theta} = \text{softmax} \circ c_{\theta} \circ h_{\theta}$, with h_{θ} and c_{θ} denoting the feature extractor and classifier, respectively. Given an unlabeled test stream $\{x_t\}_{t=1}^T$, each $x_t = \{x_{t,i}\}_{i=1}^{N_t}$ is a mini-batch sampled from $\mathcal{P}_t(X)$. The target distribution may evolve over time under temporally correlated covariate and class-prior shifts, where $\pi_t^Y(\cdot) := \mathcal{P}_t(Y = \cdot)$. We write $\theta_t = \{\theta_t^l\}_{l=1}^L$ for the trainable parameters at time step t , partitioned into L layer groups.

At each step, online TTA updates parameters θ_t using only the current unlabeled data x_t :

$$\theta_{t+1}^l = \theta_t^l - \eta \nabla_{\theta_t^l} \mathcal{L}_{\text{TTA}}(x_t; \theta_t), \quad (1)$$

where $\mathcal{L}_{\text{TTA}}$ is typically self-supervised. However, when test streams exhibit strong temporal correlation, such homogeneous and unconstrained updates allow biased gradients to accumulate unchecked across network layers.

B. Decoupled Control View of DCF

We formulate DCF as a decoupled online control framework for temporally correlated streams. DCF follows the route–adapt–retain paradigm: 1) *route* instances into trusted and routed-away subsets; 2) *adapt* via subset-specific objectives, reusing routed-away instances for geometry repair; and 3) *retain* candidate updates according to layer-wise source compatibility.

For x_t with $\mathcal{R}_t \neq \emptyset$, DCF first forms a sample-side candidate update θ_t^+ :

$$\theta_t^+ = \theta_t - \eta \nabla_{\theta_t} (\mathcal{L}_{\text{trust}}(\mathcal{R}_t) + \mathcal{L}_{\text{geo}}(\mathcal{U}_t)), \quad (2)$$

where $\mathcal{R}_t$ and $\mathcal{U}_t$ are trusted and routed-away sets. $\mathcal{L}_{\text{trust}}$ enforces consistency on probe-selected evidence, while $\mathcal{L}_{\text{geo}}$ conducts Routed-away Geometry Repair (RGR). Rather than accepting θ_t^+ globally, Curvature-aware Layer Retention (CLR) selectively filters updates via layer-wise retention:

$$\theta_{t+1} = \text{Retain}_{\mu_t, I_t, I_0}(\theta_t^+, \theta_0), \quad (3)$$

which prevents updates from being indiscriminately retained in source-sensitive layers.

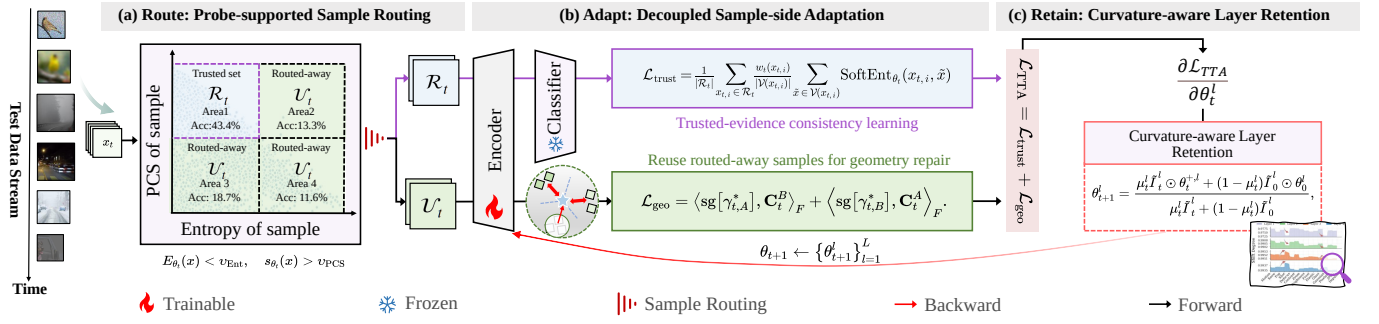


Fig. 3: **Overview of DCF.** DCF implements a route–adapt–retain pipeline. Probe-supported Sample Routing partitions instances into trusted ($\mathcal{R}_t$, Area 1) and routed-away ($\mathcal{U}_t$, Areas 2–4) sets via predictive entropy and Fourier stress response. Trusted instances supply PCS-weighted consistency supervision, while routed-away instances drive optimal-transport geometry repair via RGR. Finally, CLR selectively retains candidate parameter updates layer-by-layer via curvature-aware barycentric anchoring.

C. Route: Probe-supported Sample Routing

The first control decision is sample routing: partitioning the target batch before adaptation. Confidence alone is unreliable, as low-entropy predictions may still rely on shortcut-sensitive cues under shift (O1). We therefore pair entropy with a structured Fourier stress probe that perturbs frequency-direction patterns without spatially rearranging image content and exposes prediction responsiveness. Samples are trusted only when they are both confident and sufficiently stress-responsive, filtering out confident but stress-inert predictions.

Fourier-basis stress probe. At step t , we generate a stressed batch $x_t^s = \mathcal{A}^s(x_t)$ via an additive Fourier basis perturbation:

$$\begin{aligned} \Phi_{f,\omega}(i,j) &= R \sin(2\pi f(i \cos \omega + j \sin \omega - \pi/4)), \\ [\mathcal{A}^s(x_t)]_{i,j,c} &= \text{Clamp}_{[0,1]}([x_t]_{i,j,c} + \sigma_c \Phi_{f_c,\omega_c}(i,j)). \end{aligned} \quad (4)$$

Here, $(i,j) \in [0,1]^2$ are normalized coordinates, and R ensures $\|\Phi_{f,\omega}\|_2^2/(HW) = 1$. Per channel, independently draw $f_c \sim \mathcal{U}[1, 224]$ (nominal cycles/image), $\omega_c \sim \mathcal{U}[0, \pi]$, and $\sigma_c \sim \text{Exp}(1/\lambda)$ with rate $1/\lambda$. Thus, λ is the expected pre-clamp per-channel RMS amplitude.

For sample $x_{t,i}$, let $\hat{y}_t(x_{t,i}) = \arg \max_y p_{\theta_t}(y | x_{t,i})$. The *Perturbation Consistency Score* (PCS) is formulated as:

$$s_{\theta_t}(x_{t,i}) = [p_{\theta_t}(\hat{y}_t(x_{t,i}) | x_{t,i}) - p_{\theta_t}(\hat{y}_t(x_{t,i}) | \mathcal{A}^s(x_{t,i}))]_+. \quad (5)$$

Higher PCS indicates stronger responsiveness of the original prediction to the structured Fourier probe, whereas lower PCS indicates stress-inert behavior. Combined with predictive entropy, PCS therefore provides a complementary signal for distinguishing confident candidates according to how their predictive evidence responds to structured stress.

Theoretical interpretation. Motivated by frequency-dependent model sensitivity and texture bias [39, 40], we characterize PCS through a local probability expansion. For a fixed input x , let $\delta = \mathcal{A}^s(x) - x$, $Q_q(x) = \mathbb{E}_q[\delta \delta^\top]$, and $g_x = \nabla_x p_{\theta_t}(\hat{y}_t(x) | x)$, holding the original predicted class fixed. For a locally smooth probability function and sufficiently small perturbations, retaining the positive-part operation in Eq. (5) gives $\mathbb{E}_q[s_{\theta_t}(x)^2] = \mathbb{E}_q[(-g_x^\top \delta)_+^2] + O(\mathbb{E}_q[\|\delta\|_2^3])$. The leading term is bounded above by $g_x^\top Q_q(x) g_x$, where $Q_q(x)$ is the perturbation

second-moment matrix; no zero-mean or symmetry assumption is imposed after clamping. Thus, this relation establishes PCS as a structured directional-sensitivity measure whose response depends jointly on the prediction gradient and the directions excited by the Fourier probe. The controlled shortcut interventions in Section IV-C provide the empirical bridge to routing: among confidence-matched candidates in the evaluated settings, task-relevant predictions exhibit stronger PCS responses and are preferentially retained by the joint entropy–PCS criterion. **Sample routing.** We quantify predictive uncertainty by the entropy $E_{\theta_t}(x_{t,i}) = -\sum_{k=1}^K p_{\theta_t}(k | x_{t,i}) \ln p_{\theta_t}(k | x_{t,i})$. Using uncertainty and stress sensitivity, we partition x_t into two complementary subsets. The *trusted set* (Area 1 in Figure 3) is defined as

$$\mathcal{R}_{\theta_t}(x_t) = \{x_{t,i} \in x_t \mid E_{\theta_t}(x_{t,i}) < v_{\text{Ent}}, s_{\theta_t}(x_{t,i}) > v_{\text{PCS}}\}, \quad (6)$$

where v_{Ent} and v_{PCS} are thresholds for confidence and stress response. The routed-away set (Areas 2–4 in Figure 3) is

$$\mathcal{U}_{\theta_t}(x_t) = x_t \setminus \mathcal{R}_{\theta_t}(x_t). \quad (7)$$

As illustrated in Figure 3 (left), the two thresholds bisect the (E, s) diagnostic space into four quadrants: Area 1 ($E < v_{\text{Ent}}, s > v_{\text{PCS}}$), Area 2 ($E \geq v_{\text{Ent}}, s > v_{\text{PCS}}$), Area 3 ($E < v_{\text{Ent}}, s \leq v_{\text{PCS}}$), and Area 4 ($E \geq v_{\text{Ent}}, s \leq v_{\text{PCS}}$). For brevity, we denote the trusted and routed-away sets as $\mathcal{R}_t$ and $\mathcal{U}_t$ in subsequent sections.

D. Adapt: Decoupled Sample-side Adaptation

After sample routing, $\mathcal{R}_t$ and $\mathcal{U}_t$ receive decoupled supervisory signals. Trusted instances drive label-like consistency, while routed-away instances are repurposed for geometry repair rather than discarded or hard pseudo-labeled. The resulting sample-side objective is $\mathcal{L}_{\text{trust}} + \mathcal{L}_{\text{geo}}$, which produces the candidate update in Eqn. (2). If $|\mathcal{U}_t| = 0$, $\mathcal{L}_{\text{geo}}$ is bypassed; if $|\mathcal{R}_t| = 0$, the entire adaptation step is bypassed and prediction uses the current model.

Trusted-evidence consistency learning. For each $x_{t,i} \in \mathcal{R}_t$, we construct a multi-view set $\mathcal{V}(x_{t,i}) = \{x_{t,i}, \mathcal{A}^w(x_{t,i}), \mathcal{A}^s(x_{t,i})\}$, where $\mathcal{A}^w$ is weak augmentation [12] and $\mathcal{A}^s$ is the Fourier

stress probe (Eqn. (4)). We assign each trusted instance a normalized PCS weight:

$$w_t(x_{t,i}) = \frac{|\mathcal{R}_t| \exp\left(\frac{\tau[s_{\theta_t}(x_{t,i}) - s_{\min}]}{s_{\max} - s_{\min} + \varepsilon_w}\right)}{\sum_{x_{t,j} \in \mathcal{R}_t} \exp\left(\frac{\tau[s_{\theta_t}(x_{t,j}) - s_{\min}]}{s_{\max} - s_{\min} + \varepsilon_w}\right)}, \quad (8)$$

where τ modulates sharpness, $\varepsilon_w = 10^{-6}$, and $s_{\min}, s_{\max}$ are the PCS extrema in $\mathcal{R}_t$. With cross-entropy $\text{SoftEnt}_{\theta_t}(u, v) := -\sum_{k=1}^K p_{\theta_t}(k | u) \log p_{\theta_t}(k | v)$, the consistency loss is computed as:

$$\mathcal{L}_{\text{trust}} = \frac{1}{|\mathcal{R}_t|} \sum_{x_{t,i} \in \mathcal{R}_t} \frac{w_t(x_{t,i})}{|\mathcal{V}(x_{t,i})|} \sum_{\tilde{x} \in \mathcal{V}(x_{t,i})} \text{SoftEnt}_{\theta_t}(x_{t,i}, \tilde{x}), \quad (9)$$

with $\mathcal{L}_{\text{trust}} = 0$ if $|\mathcal{R}_t| = 0$.

Routed-away geometry repair. While trusted filtering limits unreliable pseudo-label supervision, discarding routed-away samples removes target structural context (**O2**) and may exacerbate representation bias. To exploit this context without hard pseudo-labels, DCF repurposes $\mathcal{U}_t$ through Routed-away Geometry Repair (RGR) as optimal transport (OT) alignment against source-initialized class centroids. The marginal constraints regulate aggregate class allocation, while source classifier weights provide initial semantic references without source data.

Diagnostic & Prototypes: For a feature set $\mathbf{Z} = \{\mathbf{z}_i\}_{i=1}^B$ with $B = |\mathbf{Z}|$, we assess prediction concentration via the *centroid entropy deficit* $I(\mathbf{Z}) = \ln K + \sum_{k=1}^K \bar{p}_k \ln \bar{p}_k$, where $\bar{\mathbf{z}} = \frac{1}{B} \sum_{i=1}^B \mathbf{z}_i$ and $\bar{\mathbf{p}} = \text{softmax}(c_{\theta_t}(\bar{\mathbf{z}}))$. A large deficit indicates severe prediction concentration, motivating geometric regularization. When $n_t := |\mathcal{U}_t| = 0$, we set $\mathcal{L}_{\text{geo}} = 0$ and retain $\mathbf{M}_{t+1} = \mathbf{M}_t$. Otherwise, for routed-away samples we construct two stochastic views $u_{t,i}^V = \mathcal{A}^{w_V}(u_{t,i})$ ($V \in \{A, B\}$) and extract normalized features $\mathbf{z}_{t,i}^V = h_{\theta_t}(u_{t,i}^V) / \|h_{\theta_t}(u_{t,i}^V)\|_2$. Let W_c be the c -th row of the frozen source classifier. Shared centroids start from $\mathbf{M}_1 \leftarrow \mathbf{M}_0$, with $\mathbf{M}_{0,c} = W_c / \|W_c\|_2$, providing test-time anchors without accessing source training data.

Dynamic-marginal OT: For each view, we solve an entropic OT assignment to the shared centroids under the cosine cost $[\mathbf{C}_t^V]_{ic} = 1 - \cos(\mathbf{z}_{t,i}^V, \mathbf{M}_{t,c})$ ($V \in \{A, B\}$). With uniform sample marginals $a_t = \mathbf{1}_{n_t} / n_t \in \Delta^{n_t-1}$, we allow the centroid marginal to adapt conservatively to potential changes in target class prevalence. Specifically, we first estimate the batch-level target prior from stop-gradient predictions and maintain its exponential moving average:

$$\hat{\pi}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \text{sg}[p_{\theta_t}(\cdot | x_{t,i})], \quad (10)$$

$$\bar{b}_t = (1 - \rho) \bar{b}_{t-1} + \rho \hat{\pi}_t, \quad \bar{b}_0 = \mathbf{1}_K / K, \quad (11)$$

where $\rho = 0.01$ and the uniform initialization and slow EMA suppress transient errors in $\hat{\pi}_t$ during early adaptation. We then interpolate the estimated prior with the uniform marginal:

$$b_t = \alpha_b \bar{b}_t + (1 - \alpha_b) \frac{\mathbf{1}_K}{K}, \quad (12)$$

where $\alpha_b = 0.01$ limits deviation from the uniform marginal. For $u = \mathbf{1}_K / K$, $\|b_t - u\|_1 \leq 2\alpha_b$ and $b_{t,c} \geq (1 - \alpha_b) / K$, enabling gradual adaptation without severe imbalance. The

slow EMA mitigates early fluctuations, though persistent bias may affect the estimated prior. A tiny routed-away set limits target support, while an overly dominant one reduces trusted supervision. If $\mathcal{R}_t = \emptyset$, the update is skipped and $\bar{b}_t = \bar{b}_{t-1}$. The entropic OT plans are then obtained by solving:

$$\gamma_{t,V}^* = \arg \min_{\gamma \in \Pi(a_t, b_t)} \langle \gamma, \mathbf{C}_t^V \rangle_F + \varepsilon_{\text{OT}} \sum_{i,c} \gamma_{ic} (\log \gamma_{ic} - 1), \quad (13)$$

where $\Pi(a_t, b_t) = \{\gamma \in \mathbb{R}_+^{n_t \times K} \mid \gamma \mathbf{1}_K = a_t, \gamma^\top \mathbf{1}_{n_t} = b_t\}$ and $\varepsilon_{\text{OT}} > 0$. In practice, $\gamma_{t,V}^*$ is efficiently computed via Sinkhorn–Knopp iterations and detached during backpropagation.

Cross-view alignment & Centroid update: Instead of supervising each view independently with its assignment, we enforce cross-view structural consistency over soft transport distributions:

$$\mathcal{L}_{\text{geo}} = \langle \text{sg}[\gamma_{t,A}^*], \mathbf{C}_t^B \rangle_F + \langle \text{sg}[\gamma_{t,B}^*], \mathbf{C}_t^A \rangle_F. \quad (14)$$

This objective encourages cross-view agreement in soft sample-to-prototype alignment, providing a geometric regularizer that helps preserve target structure.

Centroid Update & Drift Mitigation: Centroids are updated outside backpropagation using the accumulated OT-weighted feature statistics. Letting $m_{t,c} = \sum_{V \in \{A, B\}} \sum_{i=1}^{n_t} \text{sg}[\gamma_{t,V}^*(i, c)]$, the prototype feature mean and momentum update are:

$$\bar{\mathbf{z}}_{t,c} = \frac{1}{m_{t,c}} \sum_{V \in \{A, B\}} \sum_{i=1}^{n_t} \text{sg}[\gamma_{t,V}^*(i, c)] \mathbf{z}_{t,i}^V, \quad (15)$$

$$\mathbf{M}_{t+1,c} \leftarrow \text{norm}(\rho_M \mathbf{M}_{t,c} + (1 - \rho_M) \bar{\mathbf{z}}_{t,c}), \quad (16)$$

where $\text{norm}(v) = v / \|v\|_2$ and $\rho_M = 0.99$. Under exact OT constraints, $m_{t,c} = 2b_{t,c} > 0$. In the numerical implementation, we retain the previous centroid by setting $\bar{\mathbf{z}}_{t,c} = \mathbf{M}_{t,c}$ when the computed mass is negligible. Otherwise, the OT-weighted feature mean enters the momentum update in Eqn. (16).

E. Retain: Curvature-aware Layer Retention

The decoupled sample-side adaptation step produces a candidate update θ_t^+ retained selectively across layers. Under temporally correlated shifts, source sensitivity varies across depth (**O3**): some layers benefit from plasticity, whereas others are prone to representational drift. Accordingly, Curvature-aware Layer Retention (CLR) performs curvature-aware layer retention, preserving updates in source-compatible layers while anchoring high-mismatch layers to the source model. This yields a layer-wise proximal problem with a closed-form, proxy-weighted barycenter of the candidate and source parameters.

Curvature-proximal formulation. For each layer l , we choose θ_{t+1}^l by solving

$$\min_{\{\theta_{t+1}^l\}_{l=1}^L} \sum_{l=1}^L \left(\underbrace{\frac{\mu_t^l}{2} \|\theta_{t+1}^l - \theta_t^{+,l}\|_{I_t^l}^2}_{\text{preserve plasticity}} + \underbrace{\frac{1 - \mu_t^l}{2} \|\theta_{t+1}^l - \theta_0^l\|_{I_0^l}^2}_{\text{prevent forgetting}} \right), \quad (17)$$

where $\|v\|_I^2 := \sum_j I_j v_j^2$. Here $I_t^l, I_0^l \in \mathbb{R}_+^{d_l}$ denote non-negative diagonal curvature-aware proxy vectors for the candidate and source models, respectively, and

$$\tilde{I}_t^l := I_t^l + \varepsilon_{\text{ret}} \mathbf{1}, \quad \tilde{I}_0^l := I_0^l + \varepsilon_{\text{ret}} \mathbf{1},$$

with $\varepsilon_{\text{ret}} = 10^{-6}$. The layer-wise gate $\mu_t^l \in (0, 1]$ controls how much of the candidate update is retained. A larger μ_t^l keeps the layer closer to $\theta_t^{+,l}$, while a smaller μ_t^l pulls it closer to the source anchor θ_0^l .

Closed-form retention update. Since Eqn. (17) is separable across layers and coordinates, the solution is

$$\theta_{t+1}^l = \frac{\mu_t^l \tilde{I}_t^l \odot \theta_t^{+,l} + (1 - \mu_t^l) \tilde{I}_0^l \odot \theta_0^l}{\mu_t^l \tilde{I}_t^l + (1 - \mu_t^l) \tilde{I}_0^l}, \quad (18)$$

where $\odot$ and division are applied element-wise.

Curvature-aware proxy and retention gate. The default curvature-aware proxy for the candidate model is estimated from pseudo-label gradients on the current batch:

$$I_t^l = \frac{1}{N_t} \sum_{i=1}^{N_t} g_t^l(x_{t,i}) \odot g_t^l(x_{t,i}), \quad (19)$$

where $g_t^l(x_{t,i}) = \nabla_{\theta_t^{+,l}} \log p_{\theta_t^+}(\hat{y}_t^+(x_{t,i}) \mid x_{t,i})$ and $\hat{y}_t^+(x_{t,i}) = \arg \max_y p_{\theta_t^+}(y \mid x_{t,i})$. Thus, I_t^l averages element-wise squared per-sample gradients. The source proxy I_0^l is computed analogously at θ_0 on the same unlabeled batch x_t with the time index omitted for brevity. The layer-wise retention gate is

$$\mu_t^l = \mu_0 \exp\left(-\frac{\|\tilde{I}_t^l - \tilde{I}_0^l\|_2^2}{d_l}\right), \quad (20)$$

where $d_l = \dim(\theta^l)$, and μ_0 is the maximum retention gate. Similar curvature-aware proxy profiles yield a large μ_t^l and retain the candidate update, whereas mismatch lowers μ_t^l and anchors the layer toward the source model. Thus, CLR preserves safe plasticity while suppressing source-sensitive drift.

IV. EXPERIMENTS

A. Setup

Benchmarks. We evaluate DCF on five widely used test-time robustness benchmarks covering both synthetic corruptions and natural distribution shifts. For synthetic shifts, we use ImageNet-C (IN-C) [41], ImageNet-3DCC (IN-3DCC) [42], and ImageNet-C-Bar (IN-C-Bar) [43], which provide diverse corruption patterns for controlled evaluation under covariate shifts. For natural shifts, we consider ImageNet-R (IN-R) [44] and ImageNet-Sketch (IN-Sketch) [45], which capture variations in rendering style, texture, and visual abstraction. To further assess cross-domain transfer, we follow MME [46] and use DomainNet-126, a 126-class subset of DomainNet [47] spanning four domains: Clipart, Painting, Real, and Sketch. Together, these benchmarks provide a comprehensive evaluation of TTA across diverse distribution shifts.

Baselines. We compare DCF against the frozen source model (No Adapt) and representative source-free TTA baselines across diverse paradigms: normalization adaptation (BN Adapt [27]), entropy minimization (Tent [6]), continual/dynamic TTA via self-training (CoTTA [12]), memory bank (RoTTA [13]), or robust normalization (TRIBE [15]), self-bootstrapping adaptation (SPA [10], PTTA [11]), sharpness-aware recovery (SAR [9]), disentangled-factor filtering (DeYO [35]), and energy alignment (AEA [14]).

Algorithm 1: Proposed DCF algorithm

Require: Source parameters θ_0 , learning rate η , thresholds $v_{\text{Ent}}, v_{\text{PCS}}$
Input: Unlabeled test stream $\{x_t\}_{t=1}^T$
1 Initialize $\theta_1 \leftarrow \theta_0$, $\mathbf{M}_1 \leftarrow \mathbf{M}_0$, and $\bar{b}_0 \leftarrow \mathbf{1}_K/K$;
2 **for** each mini-batch x_t at time $t = 1, 2, \dots, T$ **do**
 // Stage 1: Route: Probe-supported
 Sample Routing
3 Initialize $\mathcal{R}_t \leftarrow \emptyset$, $\mathcal{U}_t \leftarrow \emptyset$;
4 **for** each sample $x_{t,i} \in x_t$ **do**
5 Compute predictive entropy $E_{\theta_t}(x_{t,i})$;
6 Compute PCS score $s_{\theta_t}(x_{t,i})$ via Eqn. (5);
7 **if** $E_{\theta_t}(x_{t,i}) < v_{\text{Ent}}$ **and** $s_{\theta_t}(x_{t,i}) > v_{\text{PCS}}$ **then**
8 $\mathcal{R}_t \leftarrow \mathcal{R}_t \cup \{x_{t,i}\}$;
9 **else**
10 $\mathcal{U}_t \leftarrow \mathcal{U}_t \cup \{x_{t,i}\}$;
11 **if** $\mathcal{R}_t = \emptyset$ **then**
12 $(\theta_{t+1}, \mathbf{M}_{t+1}, \bar{b}_t) \leftarrow (\theta_t, \mathbf{M}_t, \bar{b}_{t-1})$;
13 **continue**
 // Stage 2: Adapt: Decoupled
 Sample-side Adaptation
14 $\mathcal{L}_{\text{trust}}, \mathcal{L}_{\text{geo}} \leftarrow 0$, $\mathbf{M}_{t+1} \leftarrow \mathbf{M}_t$;
15 Update $\bar{b}_t$ using $\hat{\pi}_t$ via Eqs. (10) and (11);
16 Compute $\mathcal{L}_{\text{trust}}$ on $\mathcal{R}_t$ via Eqn. (9);
17 **if** $\mathcal{U}_t \neq \emptyset$ **then**
18 Compute b_t and OT plans $\gamma_{t,A}^*, \gamma_{t,B}^*$ via
19 Eqs. (12) and (13);
20 Compute $\mathcal{L}_{\text{geo}}$ and update $\mathbf{M}_{t+1}$ via Eqs. (14)
21 and (16);
22 $\theta_t^+ \leftarrow \theta_t - \eta \nabla_{\theta_t} (\mathcal{L}_{\text{trust}} + \mathcal{L}_{\text{geo}})$;
 // Stage 3: Retain: Curvature-aware
 Layer Retention
23 **for** $l = 1, \dots, L$ **do**
24 Estimate curvature-aware proxies I_t^l, I_0^l via
25 Eqn. (19);
26 Compute retention gate μ_t^l via Eqn. (20);
27 Update layer θ_{t+1}^l via Eqn. (18);
Output: Adapted model parameters θ_{T+1}

All baselines use official implementations and default hyperparameters unless specified otherwise.

Models and Implementation Details. Following prior TTA works [12, 13, 15], we use pre-trained ResNet-50 [1] backbones from RobustBench [48] and Torchvision, with batch size 64. Following Tent [6], SGD optimizes only normalization affine parameters (learning rate 2.5×10^{-4} , momentum 0.9). For CLR, block groups correspond to ResNet layer blocks (stem BN (initial_bn) and layer1-layer4) or ViT encoder blocks. For DCF, hyperparameters ($v_{\text{PCS}}, v_{\text{Ent}}, \mu_0, \tau, \lambda$) are set to (0.2, 0.6 $\ln K$, 0.99, 1.5, 0.2). The optimal transport problem in Eqn. (13) is solved via Sinkhorn-Knopp iterations with entropic regularization $\varepsilon_{\text{OT}} = 0.5$, and $N_{\text{sk}} = 3$ iterations. All results are averaged over 5 runs with random stream orderings on a single NVIDIA A100 GPU.

Scenarios. To rigorously evaluate robustness and adaptability under temporally correlated, non-stationary image streams, we design three synthetic scenarios simulating evolving corruptions (illustrated in Figure 4). Let $\rightarrow$ and $\rightsquigarrow$ denote covariate and joint covariate-label shifts, respectively:

- **Temporally correlated Covariate Shifts (T-CS):** Corruptions shift sequentially at a fixed severity level, denoted as $\mathcal{P}_1^{C_1} \rightarrow \mathcal{P}_2^{C_2} \rightarrow \dots \rightarrow \mathcal{P}_t^{C_t}$, mimicking continuous natural environmental changes (e.g., motion blur $\rightarrow$ snow $\rightarrow$ fog).
- **Temporally correlated Covariate and Label Shifts (T-CS-LS):** Covariate shifts are coupled with dynamic class imbalance: $(\mathcal{P}_1^{C_1}, \pi_1^Y) \rightsquigarrow (\mathcal{P}_2^{C_2}, \pi_2^Y) \rightsquigarrow \dots \rightsquigarrow (\mathcal{P}_t^{C_t}, \pi_t^Y)$. Following [13], the time-varying label distribution π_t^Y is sampled from a Dirichlet distribution with parameter $\delta = 0.1$.
- **Temporally correlated Covariate Shifts with Mixed Severity Levels (T-CS-MSL):** Covariate shifts with randomly varying intensities, $\mathcal{P}_1^{C_1, \sigma_1} \rightarrow \mathcal{P}_2^{C_2, \sigma_2} \rightarrow \dots \rightarrow \mathcal{P}_t^{C_t, \sigma_t}$, where severity $\sigma_t \sim \text{Unif}\{1, \dots, 5\}$, evaluating robustness across mild-to-severe domain changes.

To evaluate long-horizon stability, we include a **Long-horizon Temporally Correlated** test stream extended from the continually evolving setting in [12]. We formulate this extended sequence as $\mathcal{S}_{\text{long}} = \bigoplus_{r=1}^{15} (\mathcal{P}_{r,1} \rightarrow \dots \rightarrow \mathcal{P}_{r,15})$, exposing models to 1,125,000 samples sequentially across 225 dynamic corruptions on ImageNet-C. We set the adaptation rounds to 15, which is sufficient to reveal the tendency of baseline methods to collapse over time.

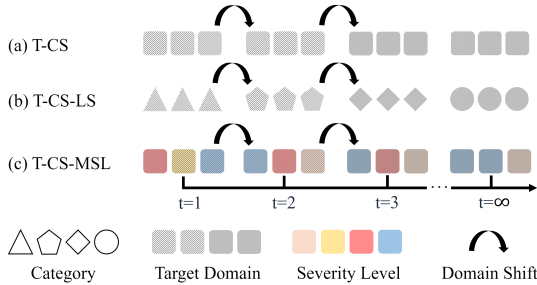


Fig. 4: **Illustration of temporally correlated test streams.** T-CS, T-CS-LS, and T-CS-MSL denote covariate shifts, added label shifts, and mixed severities, respectively.

B. Main Results

Robustness on Temporally Correlated Shifts. As shown in Table I, DCF achieves state-of-the-art accuracy across synthetic (T-CS, T-CS-LS, T-CS-MSL) and natural shift (IN-R, IN-Sketch) benchmarks. On *synthetic corruptions* (Table I, left), it obtains the highest T-CS average accuracy of 43.96%, outperforming DeYO (42.35%) and SAR (40.25%), achieving 45.71% accuracy on IN-C-Bar. Under Dirichlet label shifts (T-CS-LS) and dynamically mixed severities (T-CS-MSL), DCF remains superior with 43.84% and 55.27% average accuracies. On natural shifts (Table I, right), it averages 41.58% on IN-R and IN-Sketch, exceeding the second-best DeYO (39.00%) by +2.58 pp. These results show that decoupled sample routing, subset-specific adaptation, and layer-wise stabilization effectively mitigate error accumulation in non-stationary streams.

Stability on Long-Horizon Temporally Correlated Streams.

To evaluate stability under prolonged deployment, we track adaptation across 15 rounds spanning 225 sequentially evolving domains ($> 1.12 \times 10^6$ samples) on ImageNet-C. As shown in Table IIa, most TTA baselines suffer severe performance collapse over prolonged adaptation: DeYO (40.97% at Round 1) and Tent (37.55%) fall to near-zero accuracy by Round 15 (0.12% and 0.49%, yielding $\Delta = -40.85$ pp and -37.06 pp), while RoTTA, AEA, SPA, and PTTA show similarly large regressions ($\Delta \leq -22.02$ pp). Although continual baselines such as CoTTA and SAR avoid total breakdown ($\Delta > 0$), their average accuracies remain limited ($\leq 38.90\%$). In contrast, DCF remains robust across all rounds without catastrophic collapse, achieving the highest average accuracy and positive net progression under T-CS (43.48%), T-CS-LS (43.27%), and T-CS-MSL (54.53%). These findings support our central argument that coordinated control of input evidence routing and layer-wise parameter retention improves long-horizon stability under the evaluated temporally correlated streams.

C. Ablations and Further Analysis

Effect of Components. To isolate each module’s contribution in DCF, we conduct an ablation study (Table III). The unconstrained baseline (Tent) suffers severe error accumulation, falling to 0.49% at Round 15 (Avg. 31.53%). Adding PSR improves average accuracy to 37.25% by restricting direct consistency learning to confident, probe-supported samples, and RGR further increases it to 40.55% by reusing routed-away samples for geometry repair. Yet sample-side control alone still degrades over long streams, reaching only 26.25% at Round 15. In contrast, CLR suppresses harmful layer-wise drift, maintaining 43.84% at Round 15 and achieving the best overall average of 46.27% across all scenarios. Leave-one-out results (41.43% without PSR; 45.36% without RGR) further confirm that the sample- and layer-level controls are complementary.

Analysis of Probe-Supported Sample Routing (PSR). A central premise of DCF is that low predictive entropy does not guarantee reliable adaptation evidence, since confident predictions under continuous shifts may rely on brittle shortcut cues. We evaluate PSR using three complementary analyses: controlled shortcut diagnostics, probe and routing comparisons, and selected-sample purity and separation.

(i) *Controlled Shortcut Diagnostics:* We construct a controlled 224×224 synthetic benchmark combining circle/square shapes with four sinusoidal textures as explicit shortcuts to diagnose probe sensitivity. As reported in Table IV, across nominal frequency bands, the Fourier probe preserves shape decisions ($\geq 99.4\%$ agreement) while separating shape from texture far more effectively than energy-matched pixel noise ($> 3.4 \times$ PCS ratio vs. $1.14 \times$). Across preserved, randomized, or reversed correlations, paired counterfactuals show that high-PCS samples exhibit lower shortcut dependence among confidence-matched candidates. Specifically, PCS discriminates task-relevant from shortcut-driven predictions with 0.81 AUROC (vs. 0.71 for pixel noise) and reduces trusted-set shortcut contamination from 24.6% to 11.8% at matched coverage. Additionally, on binary Colored-MNIST [49] with re-

TABLE I: Comparisons with SOTAs w.r.t. average accuracy (%) across all domains within each dataset. Left: T-CS, T-CS-LS, and T-CS-MSL scenarios (IN-C, IN-3DCC, and IN-C-Bar). Right: Natural shifts (IN-R and IN-Sketch). Best results are **bolded**.

Methods	T-CS				T-CS-LS				T-CS-MSL				Natural Shift		
	IN-C	IN-3DCC	IN-C-Bar	Avg.	IN-C	IN-3DCC	IN-C-Bar	Avg.	IN-C	IN-3DCC	IN-C-Bar	Avg.	IN-R	IN-Sketch	Avg.
No Adapt	7.19	14.35	10.75	10.76	7.19	14.37	10.75	10.77	18.27	24.75	20.83	21.28	36.39	14.61	25.50
BN Adapt ICML 15	29.83	34.61	39.02	34.49	29.66	34.55	39.12	34.44	45.45	49.30	50.71	48.49	39.66	25.29	32.48
Tent ICLR 21	37.55	39.47	43.03	40.02	37.30	39.17	42.89	39.79	50.76	52.48	53.75	52.33	42.82	30.51	36.67
CoTTA CVPR 22	31.77	36.08	40.13	35.99	31.58	36.06	40.16	35.93	47.27	50.41	51.65	49.78	40.25	26.30	33.28
RoTTA CVPR 23	31.28	36.24	37.66	35.06	31.55	36.01	37.83	35.13	47.38	50.27	51.12	49.59	39.63	28.12	33.88
SAR ICLR 23	38.26	39.47	43.01	40.25	38.17	39.44	42.92	40.18	50.90	52.62	53.75	52.42	42.99	31.89	37.44
DeYO ICLR 24	40.97	41.31	44.78	42.35	40.71	41.03	44.49	42.08	53.00	54.03	55.46	54.16	45.22	32.78	39.00
AEA ICLR 25	22.29	34.36	41.77	32.81	24.19	35.68	42.52	34.13	50.64	50.95	53.72	51.77	44.58	20.58	32.58
TRIBE AAAI 24	36.30	39.88	36.62	37.60	36.50	40.17	40.72	39.13	49.87	53.06	51.96	51.63	39.10	28.01	33.56
LAW WACV 24	33.30	37.12	41.46	37.29	35.84	38.49	42.76	39.03	50.22	52.46	53.49	52.06	42.44	29.92	36.18
SPA ICML 25	36.57	37.35	41.38	38.43	36.57	37.39	40.90	38.29	49.22	50.77	51.90	50.63	41.01	16.09	28.55
PTTA ICML 25	38.33	38.68	43.56	40.19	38.14	38.89	43.73	40.25	50.13	51.56	53.52	51.74	42.09	30.78	36.44
Ours	43.49	42.67	45.71	43.96	43.29	42.77	45.45	43.84	54.47	55.23	56.11	55.27	47.60	35.56	41.58

TABLE II: Comparison with state-of-the-art methods on IN-C under long-horizon T-CS, T-CS-LS, and T-CS-MSL w.r.t. average accuracy (%) at Round 1/7/15 and $\Delta = R_{15} - R_1$. Best results are **bolded**. Δ values are colored red if negative.

Method	(a) Long-horizon T-CS					(b) Long-horizon T-CS-LS					(c) Long-horizon T-CS-MSL				
	Time	→				Time	→				Time	→			
	R1	R7	R15	Avg.	Δ	R1	R7	R15	Avg.	Δ	R1	R7	R15	Avg.	Δ
No Adapt	7.19	7.19	7.19	7.19	0.00	7.19	7.19	7.19	7.19	0.00	18.27	18.02	18.28	18.19	0.01
BN Adapt ICML 15	29.83	29.83	29.83	29.83	0.00	29.66	29.66	29.66	29.66	0.00	45.45	45.39	45.66	45.50	0.21
Tent ICLR 21	37.55	9.32	0.49	15.79	-37.06	37.30	26.04	12.64	25.33	-24.66	50.76	43.99	7.11	33.95	-43.65
CoTTA CVPR 22	31.77	37.43	34.43	34.54	2.66	31.58	36.16	36.28	34.67	4.70	47.27	51.27	50.00	49.51	2.73
RoTTA CVPR 23	31.28	16.53	1.35	16.39	-29.93	31.55	30.83	16.56	26.31	-14.99	47.38	15.35	0.27	21.00	-47.11
SAR ICLR 23	38.26	39.99	38.45	38.90	0.19	38.17	40.13	39.31	39.20	1.14	50.90	53.63	53.80	52.78	2.90
DeYO ICLR 24	40.97	31.46	0.12	24.18	-40.85	40.71	33.43	15.84	29.99	-24.87	53.00	53.12	53.10	53.07	0.10
AEA ICLR 25	22.29	0.23	0.27	7.60	-22.02	24.19	3.82	1.87	9.96	-22.32	50.64	0.18	0.13	16.98	-50.51
TRIBE AAAI 24	36.30	19.40	19.51	25.07	-16.79	36.50	19.98	18.95	25.14	-17.55	49.87	42.62	42.27	44.92	-7.60
LAW WACV 24	33.30	32.46	13.87	26.54	-19.43	35.84	39.64	38.06	37.85	2.22	50.22	51.82	49.44	50.49	-0.78
SPA ICML 25	36.57	0.50	0.42	12.50	-36.15	36.57	10.31	4.97	17.28	-31.60	49.22	0.23	0.22	16.56	-49.00
PTTA ICML 25	38.33	29.06	0.51	22.63	-37.82	38.14	31.96	15.14	28.41	-23.00	50.13	50.40	50.55	50.36	0.42
Ours	43.49	43.11	43.84	43.48	0.35	43.29	43.14	43.37	43.27	0.08	54.47	54.47	54.65	54.53	0.18

TABLE III: Component ablation of DCF on ImageNet-C. Here, PSR, RGR, and CLR correspond to Eqs. (6) and (7), Eqn. (14), and Eqn. (18), respectively.

Components			Scenarios				Avg.
PSR	RGR	CLR	T-CS	T-CS-LS	T-CS-MSL	Long-horizon T-CS (Final Acc.)	
×	×	×	37.55	37.30	50.76	0.49	31.53
✓	×	×	39.43	40.20	52.40	16.96	37.25
✓	✓	×	41.37	41.13	53.45	26.25	40.55
×	✓	✓	40.27	40.96	52.44	32.03	41.43
✓	×	✓	41.87	42.71	53.97	42.88	45.36
✓	✓	✓	43.49	43.29	54.47	43.84	46.27

versed shortcuts, DCF achieves 88.91% accuracy, outperforming baselines (see Table V). Together, these results support the joint use of predictive entropy and PCS for routing, showing that joint confidence- and stress-responsiveness routing better isolates reliable adaptation evidence from shortcut-driven predictions.

(ii) *Stress Probe and Router Design Choices*: We next examine these choices on ImageNet-C (Figure 5). The Fourier-based stress probe achieves the highest accuracy, outperforming pixel shuffling, patch shuffling, and low-pass filtering. As shown

TABLE IV: Factorized response diagnostics of the Fourier probe. Shape agreement measures consistency between original and stressed shape predictions, while PCS measures the response of the corresponding factor-specific readout. The shape/texture PCS ratio characterizes the probe’s differential sensitivity across visual factors. Lower, middle, and upper nominal bands correspond to [1, 75), [75, 150), and [150, 224] cycles/image, respectively.

Probe	Shape agr. (%)	PCS _{shape} ($\times 10^{-3}$)	PCS _{texture} ($\times 10^{-3}$)	Shape/texture PCS ratio
Lower nominal band	99.40	241.0	69.0	3.49×
Middle nominal band	99.60	224.0	62.0	3.61×
Upper nominal band	99.70	217.0	58.0	3.74×
Energy-matched pixel noise	99.80	112.0	98.0	1.14×
DCF random Fourier	99.60	229.0	64.0	3.58×

in Figure 6, naive spatial perturbations disrupt global object layout, whereas the Fourier probe injects structured frequency-direction stress without rearranging image coordinates. Its routing value is reflected in both downstream accuracy and selected-sample quality. The coupled Entropy+PCS rule reaches

TABLE V: **Controlled color shortcut on binary Colored-MNIST.** Digit identity defines the semantic label, while color serves as a controlled spurious cue. Training labels contain 25% noise, whereas test labels are clean, with approximately 30% color-label agreement.

Method	Source	BN	Tent	CoTTA	RoTTA	SAR	DeYO
Acc.	50.93	47.21	44.36	47.21	50.23	47.21	62.37
Method	AEA	TRIBE	LAW	SPA	PTTA	DCF (Ours)	
Acc.	43.18	33.93	42.73	39.26	50.81	88.91	

43.49%, outperforming PCS-only routing and the evaluated SAR-, DeYO-, and ETAGE-style filters [9, 16, 35]. These results highlight complementary roles for confidence and stress responsiveness: entropy suppresses ambiguous predictions, whereas PCS further distinguishes confident candidates according to their structured response. Their combination therefore yields more reliable routing under the evaluated shifts.

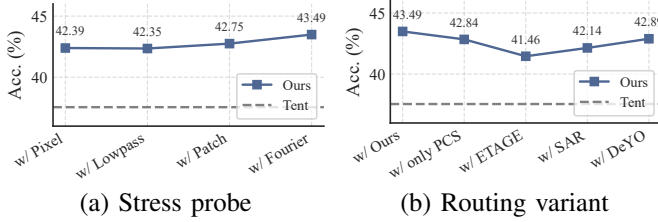


Fig. 5: **Ablation of design choices in stress probe and router on ImageNet-C (T-CS).** Our Fourier probing and entropy-PCS routing perform best, showing that structured frequency cues and joint criteria provide more reliable adaptation signals.

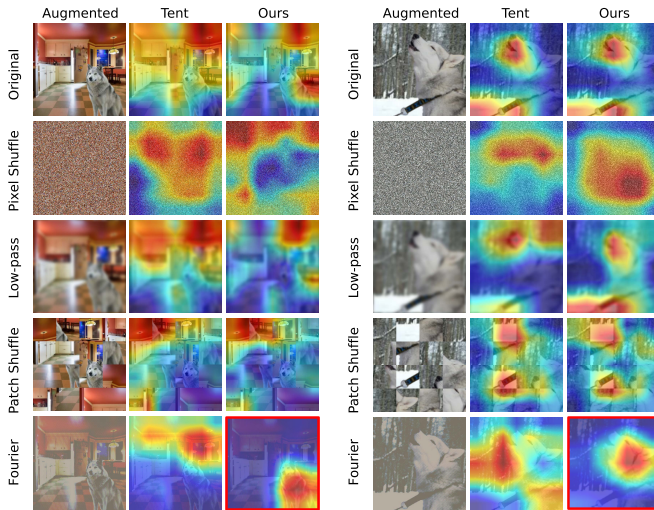


Fig. 6: **Qualitative comparison of model responses to stress probes.** The two panels show different image examples. Each compares augmented inputs and the responses of Tent and DCF under the same stress probes.

(iii) *Pseudo-Label Purity and Feature Space Separation:* Finally, we evaluate whether the gain in selected-sample quality persists independently of selection quantity on ImageNet-C. As shown in Table VI, using the same frozen model state

and matched trusted-set coverage, Entropy+PCS achieves the highest pseudo-label purity of 49.9%, compared with 44.0% for entropy-only routing and 47.1% for PCS-only routing. The diagnostic PCS-Entropy space in Figure 7 provides a consistent view: Area 3 collects confident but stress-inert candidates, whereas Area 1 contains low-entropy, probe-responsive samples and remains relatively more reliable throughout adaptation. Together with the controlled shortcut diagnostics above, these results show that entropy and PCS provide complementary routing signals and improve selected-sample reliability at matched coverage, rather than merely changing the number of selected samples.

TABLE VI: **Selected-sample quality on ImageNet-C (T-CS) at matched trusted-set coverage.** Pseudo-label purity measures correctness of selected pseudo-labels.

Method	Pseudo-label Purity (%)
Tent	42.7
+ Entropy Filter	44.0
+ PCS Filter	47.1
+ Entropy & PCS	49.9

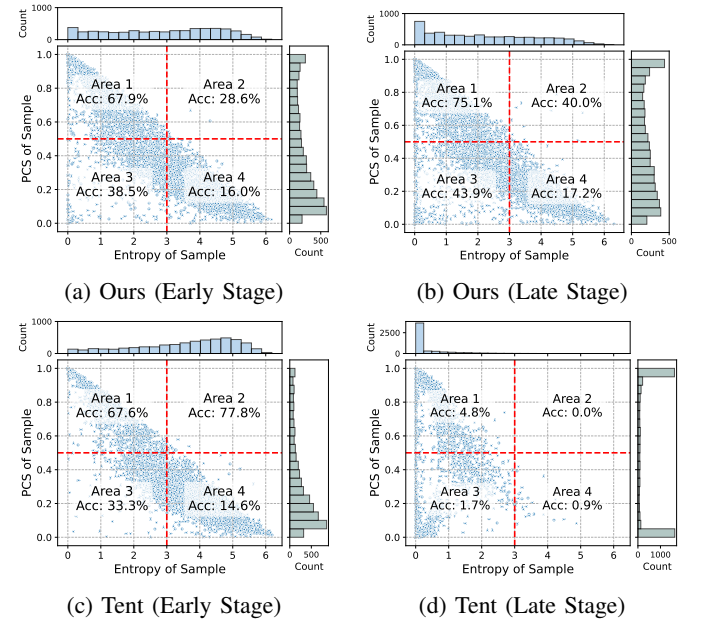


Fig. 7: **PCS-Entropy distribution of samples (ImageNet-C, long-horizon T-CS).** For DCF, the low-entropy, high-PCS diagnostic region (Area 1) remains reliable throughout adaptation, whereas Tent degrades markedly across all regions at the late stage. This comparison highlights the complementary roles of entropy and PCS in identifying reliable adaptation evidence and the superior long-horizon stability of DCF.

Analysis of Routed-away Geometry Repair (RGR). Beyond filtering unreliable samples via PSR, we investigate whether the routed-away subset $\mathcal{U}_t$ should be discarded or reused for geometry repair. To assess the role of OT in RGR, we compare it with two alternatives: *moment matching*, which aligns feature statistics with prototypes, and *prototype contrastive alignment*,

which pulls features toward nearest prototypes. As shown in Table VII, both alternatives perform worse than simply discarding $\mathcal{U}_t$ under long-horizon adaptation (41.51% and 42.33% vs. 42.88%), whereas OT-based repair improves accuracy to 42.91% with a uniform prior and 43.84% with the dynamic prior. This highlights the importance of transport-based alignment and adaptive marginal estimation. We further examine sensitivity to the routed-away proportion by varying the fixed ratio from 10% to 90% (Table VIII). Performance remains stable when 10–50% of samples are routed away, with accuracy ranging from 42.98% to 43.49% and limited centroid drift. Accuracy declines only when routed-away samples dominate the batch, reaching 22.12% at 90%, where trusted supervision becomes insufficient. Overall, these results delineate a broad stable regime, highlighting the complementarity between trusted consistency adaptation and routed-away geometry repair.

TABLE VII: **Comparison of alternative treatments for routed-away samples** on ImageNet-C. All variants share the same PSR and CLR.

Treatment of $\mathcal{U}_t$	T-CS	T-CS-LS	Long-horizon T-CS (Final Acc.)	Avg.
Discard $\mathcal{U}_t$ (Trusted-only)	41.87	42.71	42.88	42.49
Moment matching	42.81	42.82	41.51	42.38
Contrastive alignment	42.83	42.79	42.33	42.65
RGR w/ fixed uniform prior	43.01	42.80	42.91	42.91
RGR w/ dynamic prior	43.49	43.29	43.84	43.54

TABLE VIII: **Routing-ratio stress test** on ImageNet-C (T-CS). Controlled ratios are imposed per mini-batch using PSR-based ranking; *Native* uses the original routing. Acc. denotes average top-1 accuracy, and centroid drift denotes the mean ℓ_2 prototype displacement from the initial online reference.

Routed-away ratio	Acc.	Centroid drift
Native	43.49	0.0518
10%	42.98	0.0052
25%	43.49	0.0217
50%	43.18	0.0829
75%	41.64	0.1268
90%	22.12	0.0468

Analysis of Curvature-aware Layer Retention (CLR). Tracking layer-wise parameter drift in Figure 8 shows that DCF follows a “pulse-and-recover” pattern rather than persistent drift accumulation. As shown in Figure 9, the last ResNet block (*layer4*) retains the largest gate, $\mu_t^l \approx 0.98$, while *initial_bn* and *layer1* vary within 0.93–0.96, and *layer2* remains near 0.95–0.96. *Layer3* is similar but can fall to about 0.92 under strong mismatch. Responses are corruption dependent: *Motion* most strongly suppresses *initial_bn*, *Glass* affects *layer1*, and *Frost* most strongly reduces *layer3*, whereas *layer4* stays near 0.98. Temporally, suppression around domain indices 30–45 recovers toward 0.95–0.96 in later domains, indicating that CLR applies transient source anchoring rather than progressively freezing layers. Ablations in Table IX support layer-block control and compare the default gradient-square proxy with expected Fisher [51] and a Hutchinson-based diagonal second-order approximation [50], which yield small gains at additional computational cost.

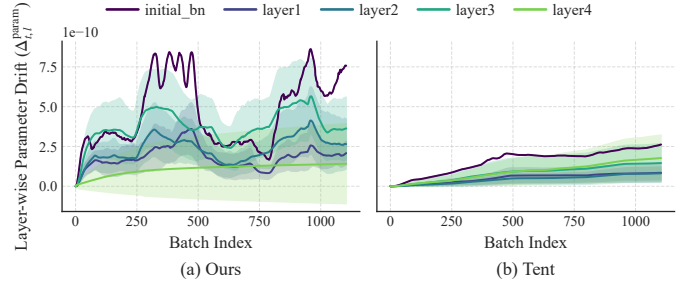


Fig. 8: **Layer-wise parameter drift on ImageNet-C (T-CS, ResNet-50)**, measured by $\Delta_{t,l}^{\text{param}} = \|\theta_t^l - \theta_0^l\|_2^2 / d_l$ (averaged per layer block). DCF exhibits a pulse-and-recover behavior, avoiding the persistent drift accumulation observed in Tent.

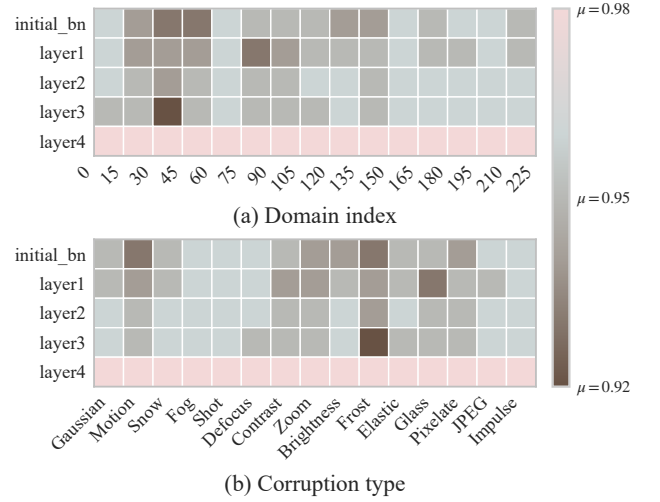


Fig. 9: **Per-layer retention gates μ_t^l of CLR (ResNet-50)**. (a) Gates over the long-horizon stream (Domain index $0 \rightarrow 225$). (b) Gates per corruption type on T-CS.

TABLE IX: **Ablation of design choices in CLR on ImageNet-C (long-horizon T-CS)**. We compare different control granularities and curvature-aware proxy estimators.

	Avg. Acc.	extra ms/img
<i>(a) Control granularity</i>		
Per-normalization-layer	42.92	0
Per-layer block (default)	43.48	0
<i>(b) Curvature-aware proxy estimator</i>		
Per-sample gradient-square (default)	43.48	0
Expected Fisher [51]	43.57	1.9
Hutchinson-based diagonal approximation	43.70	6.8

Static vs. Temporally Correlated Streams. To examine performance across streaming protocols, we compare adaptation under independently sampled (Static) and temporally correlated long-horizon (Long-horizon T-CS) test streams on ImageNet-C. As shown in Figure 10, several baselines that perform reasonably well in the Static setting, including Tent, DeYO, and AEA, degrade sharply or collapse under Long-horizon T-CS, indicating strong susceptibility to accumulated adaptation errors. In contrast, DCF achieves the highest accuracy in both settings and remains stable under

temporally correlated shifts. These results demonstrate that its sample- and layer-aware control effectively mitigates error accumulation in non-stationary continuous streams.

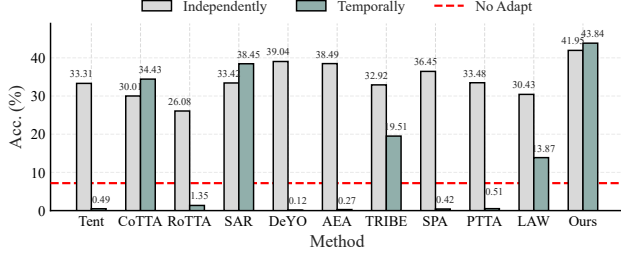


Fig. 10: **Static vs. Temporally Correlated Streams.** Comparison on ImageNet-C between independently sampled streams (Static setting) and temporally correlated long-horizon streams (Long-horizon T-CS, final round accuracy).

Generalizability across Architectures. To assess architectural generalizability, we evaluate DCF with ResNet-18 (RN-18), WideResNet-50 (WRN-50), ResNeXt-50 (RN-50), ResNet-101 (RN-101), and ViT-B/16. As shown in Table X, DCF consistently achieves the best performance across all evaluated backbones. Its effectiveness extends from lightweight ResNet-18 to deeper ResNet-101 and across distinct architectural families including WideResNet, ResNeXt, and Vision Transformer, with particularly pronounced gains on ResNet-101 and ViT-B/16, where several baselines struggle to maintain stability. These results demonstrate robust generalizability across model scales and architectures.

TABLE X: **Architectural generalizability.** DCF consistently surpasses baselines on ImageNet-C under T-CS across diverse backbones. ‘/’ indicates architectural incompatibility.

Methods	RN-18	WRN-50	RN-50	RN-101	ViT-B/16	Avg.
No Adapt	5.55	21.41	21.51	9.87	40.49	19.77
BN Adapt	25.78	33.86	32.94	32.03	40.49	33.02
Tent	31.13	43.45	42.41	41.22	45.43	40.73
CoTTA	26.97	37.27	35.56	34.86	40.32	35.00
RoTTA	24.51	37.81	36.14	35.54	/	33.50
SAR	31.74	43.48	42.88	42.70	45.42	41.24
DeYO	32.54	45.63	45.43	43.01	48.38	43.00
AEA	19.20	31.47	27.03	20.16	36.26	26.82
TRIBE	29.81	40.23	39.55	24.21	/	33.45
LAW	27.61	38.28	40.37	36.73	25.27	33.65
SPA	24.87	27.62	31.16	41.22	49.81	34.94
PTTA	31.20	42.80	40.50	41.92	47.08	40.70
Ours	34.25	47.47	46.86	47.67	54.95	46.24

Cross-Domain Transfer. Online TTA may over-specialize a model to the adaptation stream, thereby reducing its ability to generalize to unseen domains. We therefore evaluate post-adaptation transfer by adapting to a source shift and directly testing on unseen targets without further updates, using synthetic corruptions on ImageNet-C and artistic domains on DomainNet-126. On ImageNet-C (Figure 11a), DCF achieves the largest average gain over *No Adapt*, 30.30 ± 0.39 points, exceeding AEA (27.95 ± 0.05) by 2.35 points. This advantage is statistically significant across five matched runs (paired t -test, $t(4) = 14.30$, $p = 1.39 \times 10^{-4}$), whereas RoTTA exhibits negative transfer on several corruptions. On DomainNet-126 (Figure 11b), DCF also

outperforms DeYO on all Adaptation→Evaluation pairs, achieving 61.40% mean accuracy compared with 60.13% (+1.27 points), with a significant advantage across five matched runs (paired t -test, $t(4) = 12.01$, $p = 2.75 \times 10^{-4}$). Together, these results indicate that decoupled control preserves representations that remain transferable beyond the adaptation domain.

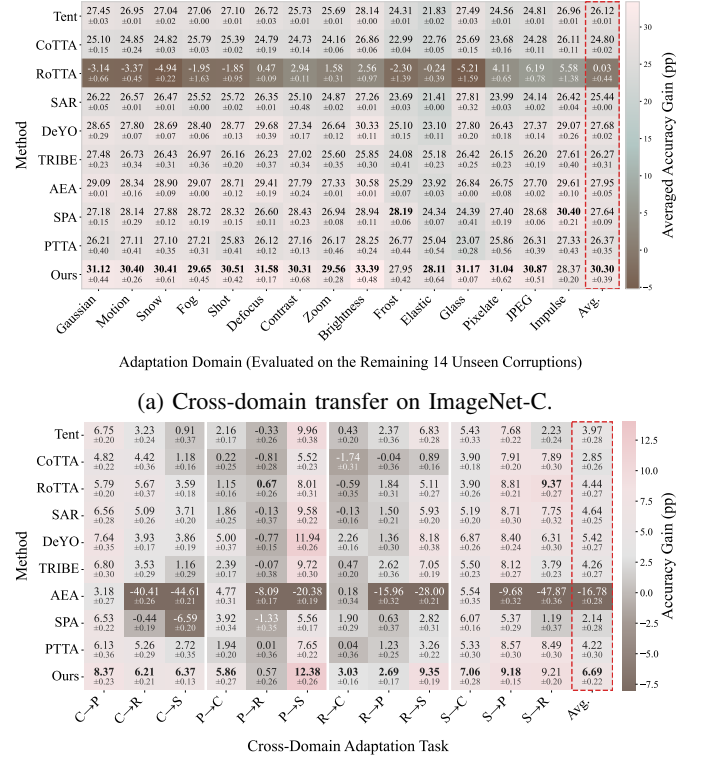


Fig. 11: **Cross-domain transfer.** (a): ImageNet-C accuracy gains (in percentage points) over *No Adapt*, averaged across all unseen corruption domains. (b): DomainNet-126 accuracy gains over *No Adapt* for each cross-domain transfer pair (Adaptation → Evaluation, where C: Clipart, P: Painting, R: Real, and S: Sketch). Results are reported as mean $\pm$ standard deviation over five runs.

Hyperparameter Sensitivity. As shown in Figure 12, DCF is generally insensitive to its key hyperparameters. In Figure 12a, the proposed dynamic CLR consistently outperforms all static retention settings across different τ and μ , highlighting the benefit of layer-wise adaptive retention. Figure 12b shows a broad high-accuracy region for (v_{PCS}, v_{Ent}) . For a new dataset, a practical rule is to start from $v_{PCS} = 0.2$ and $v_{Ent} = 0.6 \ln K$ and perform only coarse adjustment if necessary, rather than dataset-specific fine tuning. Likewise, performance varies only mildly across λ and the frequency range in Figure 12c. Finally, Figure 12d shows stable accuracy across ϵ_{OT} and N_{sk} , suggesting that reliable OT alignment can be achieved with only a few Sinkhorn iterations. Overall, DCF requires little hyperparameter tuning for stable performance.

Computational Overhead. As benchmarked on an NVIDIA A100 (Table XI), the full DCF achieves 43.48% average accuracy with 24.6 GFLOPs, 14,690 MB peak memory, and

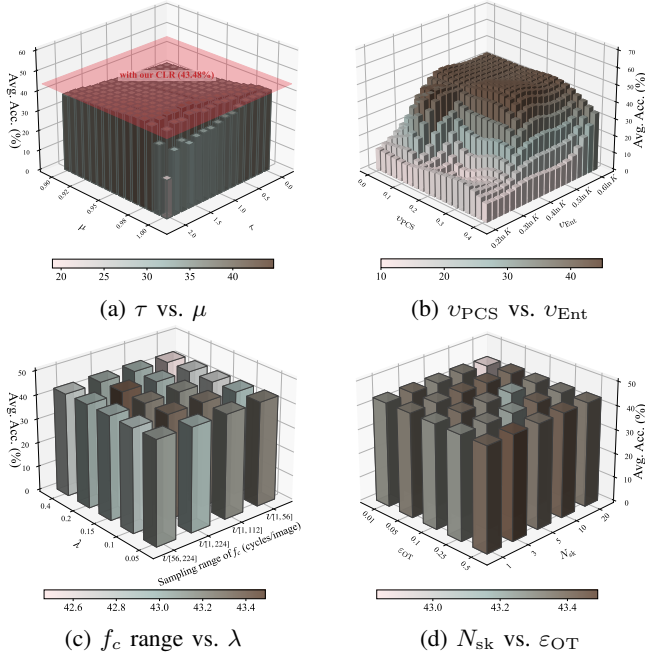


Fig. 12: **Hyperparameter sensitivity.** (a) PCS sharpness τ vs. static retention μ , where dynamic layer-wise retention (CLR, red plane) strictly surpasses static retention baselines. (b) Joint routing thresholds ($v_{\text{PCS}}, v_{\text{Ent}}$), showing a broad optimal zone balancing trusted evidence and sample volume. (c) Sensitivity to perturbation strength λ and nominal frequency sampling range f_c (cycles/image). (d) Stability across Sinkhorn iterations N_{sk} and entropic regularization ε_{OT} , confirming reliable OT alignment with low iteration counts.

5.2 ms/image latency. *DCF-Lite* reduces computational cost by removing persistent source/candidate model copies and per-sample gradient computation, retaining gradients only through the clean adaptation branch, and using a detached Fourier probe for PSR routing and transport assignment. Lite keeps the source parameter anchor as a tensor snapshot and caches the element-wise squared gradient of the first adaptation loss, capped at 10^{-4} , before the first optimizer update. This fixed reference is reused in comparisons with the current curvature proxy for retention-gate computation. With $N_{\text{sk}} = 1$, Lite requires only 9.4 GFLOPs and 6,080 MB peak memory, corresponding to reductions of 61.8% and 58.6%, respectively, relative to the full model, while reducing latency to 3.19 ms/image. It achieves 42.06% accuracy, only 1.42 points below full DCF and 3.16 points above SAR (38.90%), providing a favorable accuracy–efficiency trade-off.

V. CONCLUSIONS

Our analysis identifies a coupled sample–layer feedback process as a key mechanism of instability in temporally correlated test-time adaptation, linking heterogeneous target evidence to layer-dependent drift. To address this, we propose DCF, which decouples sample routing, subset-specific sample-side adaptation, and curvature-aware layer retention. By screening shortcut-sensitive evidence, reusing routed-away samples for optimal-transport geometry repair, and selectively

TABLE XI: **Computational overhead and lightweight variant.** Comparison of FLOPs, peak memory, per-image runtime, and average accuracy on ImageNet-C (long-horizon T-CS, ResNet-50) measured on an NVIDIA A100.

Method	FLOPs (G)	Peak mem (MB)	ms/img	Avg. Acc.
No Adapt	4.1	1834	0.5	7.19
BN Adapt	4.1	1944	0.5	29.83
Tent	8.2	5685	1.4	15.79
CoTTA	24.6	13892	2.8	34.54
SAR	16.4	6936	3.0	38.90
DeYO	10.0	5895	2.1	24.18
AEA	8.3	6936	1.6	7.60
PTTA	16.4	11872	4.1	22.63
RoTTA	28.9	12952	18.7	16.39
LAW	16.4	13364	6.9	26.54
TRIBE	65.6	18405	15.6	25.07
SPA	49.1	9635	10.3	12.50
DCF (full)	24.6	14690	5.2	43.48
DCF-Lite	9.4	6080	3.19	42.06

retaining candidate updates across layers, DCF mitigates the error-amplifying feedback loop. Extensive evaluations show that DCF consistently improves test-time robustness and avoids the late-stage collapse observed in existing TTA methods in long-horizon deployment streams.

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